

CWR in situ · Installation & user guide

Field recording of crop wild relatives (CWR) — international version

How to install the app on a phone (Android or iPhone), record a population in the field, reuse genetic-reserve polygons, and protect your data. Read it fully before your first field trip.

Part 1 · Installing the app

CWR in situ is a web application installed from a web address. You (or your coordinator) will host it and share a **URL** of the form `https://USER.github.io/florcwr-en/FlorCWR.html`. The procedure differs by phone; follow only your section.

Host this English version at its own URL, separate from any Spanish version. Two web apps served from the same path would overwrite each other's offline cache and stored records.

A. Android (Samsung, Xiaomi, Google Pixel, etc.)

1. Open the URL in **Google Chrome**.
2. Wait for the banner “✓ **available offline**”.
3. Open the Chrome menu (**three dots** ⋮) and choose “**Add to Home screen**” / “**Install app**”.
4. Open the app from the new icon and grant the **location** permission when asked.

B. iPhone (iOS)

Use Safari, not Chrome. The install and storage features only behave correctly in Safari.

1. Open the URL in **Safari** and wait for “✓ **available offline**”.
2. Tap the **Share** button (square with an upward arrow) and choose “**Add to Home Screen**”.
3. Open the app from the icon and grant the **location** permission.

Offline base map (both systems)

The app always starts offline, but the **map background** must have been loaded earlier with a connection. Before the field, with coverage, open the map and pan/zoom over your working zone so the tiles are cached for offline use.

Automatic updates

When a new version is published, the app shows an “**A new version is available — Update**” bar; tap **Update**. The first time, or after a service-worker change, you may need to force it once (hard reload, or clear site data after exporting a backup).

Part 2 · Recording a population

The form is ordered to match field workflow. It **starts with the genetic reserve**, then the population details.

Reserve block (top of the form)

- **Locality** — place name / site / municipality.
- **Genetic reserve name** — free text, e.g. “Grazalema 1”.
- **Saved reserves** — a dropdown (appears once you have saved at least one reserve) to reuse a reserve already recorded; see below.
- **Reserve area** — the reserve polygon, on the map.

Drawing the reserve polygon

Tap “Start polygon”, then build the outline in any of three ways:

1. **Tap the map** to place vertices by eye.
2. **Add GPS vertex**: stand at each corner and tap.
3. **Track on foot**: auto-record a point every few seconds while you walk the perimeter.

Use **Undo** and **Close polygon**. The area is computed in m²/ha. Tap **“Follow my position”** to show a live blue dot of your current location on the map (it stays visible even with the frequency grid on).

Reusable genetic reserves

When a **reserve name is filled in and the polygon is closed**, the reserve is **saved automatically** (name + polygon + locality) in a store separate from the population records. In a later record, pick it from **Saved reserves** and tap **Load**: the polygon, the reserve name and the locality are restored, so you don't re-survey them. The occupancy-frequency grid is **per population**, so it resets on load — define it for the new record if needed.

Population details

- **Species** (free text) and optional **Family**. If a national checklist is loaded (see below), the Species field suggests names as you type; you can still enter a name that is not on the list.
- **ID certainty**, **Observation date** (required), **Point coordinates** (GPS or typed; UTM ETRS89 shown).
- **Locality** is already filled from the reserve; **Habitat (EUNIS)**, **Substrate**, **Phenology**, **Cover**, **Herbarium voucher**, **Photos**, **Recorder**, **Notes** as applicable.

Species checklist (optional)

The international version has **no built-in species list**. Each country can load its own CWR checklist so the **Species** field offers autocomplete. In the **Data** tab, under **“Species checklist”**, load a file with one scientific name per line (CSV or TXT; a header row is detected automatically) or a JSON array of names. A counter shows how many taxa are loaded; **“Remove checklist”** returns to free-text mode. The checklist is stored on the device and travels inside the JSON backup, so a coordinator can prepare it once and share it with the team.

The checklist only assists typing — it never restricts what you can record. A name absent from the list is still saved, so unexpected finds are never blocked.

Population size estimation

Method	What it does
Total census	Type the Abundance directly and choose any Count unit.
Transect	Enter transect size (m ²) and number of individuals. Abundance = individuals × reserve-area / transect size. Count unit forced to “Individual”.
Frequency	Lay a UTM grid over the polygon; mark occupied cells (GPS or tap). Abundance = occupancy frequency (%). Count unit forced to “Frequency (%)”.

Transect note: assumes the transect density is representative of the whole reserve area (homogeneous distribution). For aggregated populations, several transects and an average are more reliable.

Occupancy frequency grid

Choose a cell size (5, 10, 25, 50, 100 m or 1 km). The grid is clipped to the polygon. Walk the population and tap **“Mark presence (GPS)”** at each occupied point; frequency = occupied cells × 100 / total cells.

Accuracy: a phone GPS is typically $\pm 4\text{--}10\text{ m}$, so $5\text{--}10\text{ m}$ cells fall below GPS accuracy; the app warns when accuracy exceeds half the cell size. Confirm fine-grid cells visually or by tapping.

Part 3 · Exporting and backing up data

Records are stored **only in the browser of the phone that took them**. The **Data** tab offers **CSV** (flat table), **GeoJSON** (points + polygons for QGIS; WGS84 with UTM ETRS89 as attributes) and **JSON backup** (full state, the only re-importable format — it also includes your saved reserves and any loaded checklist). Field names are in English.

Backup routine

After each field day, before closing the app, export a JSON backup and upload it to your shared repository.

iPhone users: the backup is critical.

Safari may erase a web app's stored data after ~7 days of inactivity. Export the JSON backup after every outing, and open the app every few days even if you record nothing.

File-name convention: `florcwr_YYYYMMDD_zone_initials.json`. Where several people record together, one person merges the individual backups in a dedicated browser, checking duplicates by species and nearby coordinates.

While the app has no shared back-end database, this manual routine keeps the team's work from being lost or overwritten. Saved reserves are shared between records on the same device, but not automatically between phones — they travel inside the JSON backup.